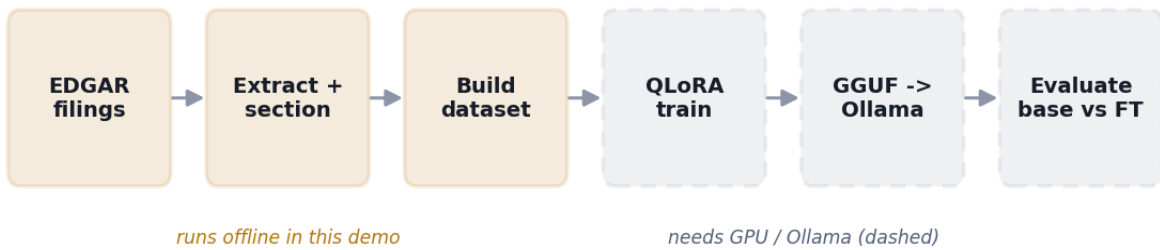
SEC 10-K → Local LLM Fine-Tuning — Pipeline Report

Extraction · dataset construction · evaluation methodology

Generated 2026-06-24 04:09 from a self-contained demo run (no network, no GPU, no Ollama).



The amber stages run end-to-end in this demo against bundled synthetic (fictional-company) 10-K filings, using the real extraction and dataset-building code. The dashed stages — QLoRA training, GGUF export, and base-vs-fine-tune evaluation — require a GPU and a running Ollama, and are documented here as methodology.

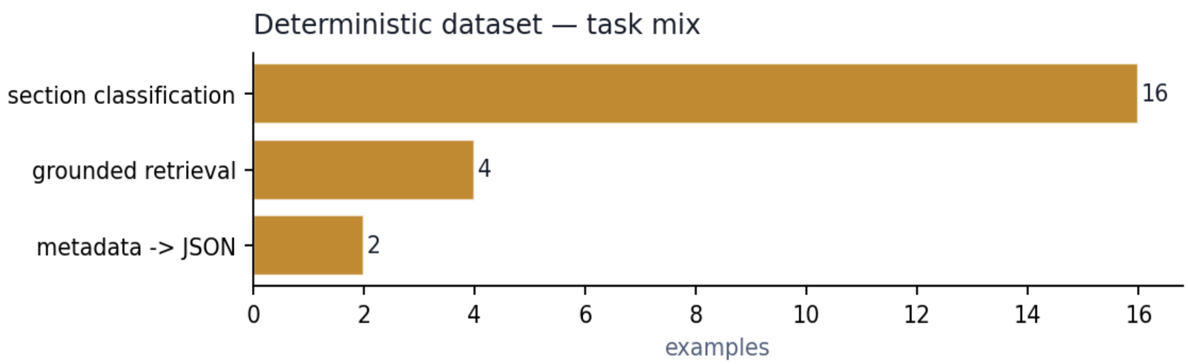
1. Extraction & sectioning

Each filing's HTML is stripped to text (inline-XBRL hidden facts removed) and split into standard Form 10-K Items. The splitter keeps the longest span per item so table-of-contents rows are discarded.

Company	Ticker	Form	Filed	Chars	Items found
Northwind Logistics Inc.	NWND	10-K	2026-02-18	4,044	8
Cascade Renewable Power Corp.	CRPC	10-K	2026-03-04	4,120	8

2. Dataset composition

The deterministic builder derives exact, programmatic labels — no teacher model required. Three task types are emitted; classification is the most abundant signal and is capped per filing to keep the mix from skewing too far toward it.



Total deterministic examples: **22** · train/eval split: **19** / **3** (chat-format JSONL).

3. Sample training examples

One example per task, as written to the JSONL (system prompt omitted for space). These are exactly what the model trains on.

section classification

> **user:** Which section of a Form 10-K does the following passage come from? Answer with the item number and title only. Passage: The Company is exposed to interest-rate risk on its variable-rate revolving credit facility and to commodity-price risk on diesel fuel. A hypothetical 100 basis-point increase in rates would not have a material effect given current borrow …
>
> **assistant:** Item 7A — Quantitative and Qualitative Disclosures About Market Risk

metadata -> JSON

> **user:** Return the following filing facts as a JSON object with keys company, ticker, form_type, filing_date, fiscal_period_end. Respond with only the JSON. company: Northwind Logistics Inc. ticker: NWND form_type: 10-K filing_date: 2026-02-18 fiscal_period_end: 2025-12-31
>
> **assistant:** {"company": "Northwind Logistics Inc.", "ticker": "NWND", "form_type": "10-K", "filing_date": "2026-02-18", "fiscal_period_end": "2025-12-31"}

grounded retrieval

> **user:** Quote the opening of the MD&A discussion from Northwind Logistics Inc.'s 10-K filed 2026-02-18, using only the filing text below. Filing text: Item 7. Management's Discussion and Analysis Revenue increased 6.2% year over year, driven by higher contracted lane pricing that more than offset a modest decline in tonnage. Operating ratio improved to 92.4% as lo …
>
> **assistant:** From Item 7 — Management's Discussion and Analysis (MD&A) of Northwind Logistics Inc.'s 10-K (filed 2026-02-18): Item 7. Management's Discussion and Analysis Revenue increased 6.2% year over year, driven by higher contracted lane pricing that more than offset a modest decline in tonnage. Operating ratio improved to 92.4% as load-optimization gains lifted trailer utilization. Liquidity remains adequate with cash and …

4. Evaluation methodology

The portfolio claim is not “the model reviews filings” but “fine-tuning measurably improves a narrow, checkable task.” The held-out eval set is scored with exact-match on section classification: the predicted Item number must equal the gold Item number. evaluate.py runs the same eval file against any Ollama model, so the stock base model and the fine-tune are compared on identical inputs.

Illustrative result shape (replace by running evaluate.py on your GPU/Ollama machine):

Model	Classification exact-match	Note
qwen2.5:7b-instruct (base)	illustrative	stock baseline
sec-analyst (QLoRA fine-tune)	illustrative	after training on this dataset

Because the labels are programmatic, the eval is fully reproducible and free of teacher-model bias — the same property that makes the deterministic tasks trustworthy training signal.

5. What runs where

Stage	Runs offline?	Requirement
Extract + section filings	yes	requests, bs4, lxml
Deterministic dataset + split	yes	stdlib only
Unit tests (21, network-free)	yes	pytest
Synthetic teacher summaries	no	local Ollama
QLoRA fine-tune + GGUF export	no	NVIDIA GPU ~10-12GB / Colab
Base-vs-fine-tune evaluation	no	Ollama + the GGUF